

AI-Based Fashion Measurement and Outfit Recommendation System

AI/ML Internship — Day 7

Name: Praneetha AS

Role: AI/ML Intern

Project Folder: Raritone

Module: Human Body Detection System

1. Introduction

This project focuses on developing an AI-powered Fashion Measurement and Outfit Recommendation System using Artificial Intelligence, Computer Vision, MediaPipe, OpenCV, and CSV Dataset Processing.

The system performs:

- human body landmark detection
- body measurement estimation
- body position analysis
- angle detection
- clothing size comparison
- intelligent outfit recommendation

The project uses AI body tracking and fashion datasets to recommend suitable clothing sizes and outfit categories.

2. Objectives of the Project

The objectives of this project are:

- Detect human body landmarks
- Estimate body measurements using AI
- Analyse body position and posture
- Detect body orientation angles
- Read clothing size data from CSV dataset
- Compare body measurements with garment measurements
- Recommend suitable outfit sizes

3. Technologies Used

Technology	Purpose
Python	Programming
OpenCV	Image Processing
MediaPipe	Human Landmark Detection
Pandas	CSV Dataset Processing
Artificial Intelligence	Recommendation Logic
Computer Vision	Pose Estimation

4. Features Implemented

The system successfully implemented:

- body landmark detection
 - skeletal visualization
 - body measurement estimation
 - body angle detection
 - body posture analysis
 - CSV dataset reading
 - clothing size prediction
 - AI outfit recommendation
-

5. Workflow of the System

Step 1 — Human Image Input

The system accepts:

- human body images
 - standing posture images
 - full body photographs
-

Step 2 — Body Landmark Detection

MediaPipe detects:

- shoulders
- elbows
- wrists
- hips
- knees
- ankles

using AI pose estimation.

Step 3 — Body Measurement Estimation

The AI system estimates:

- chest measurement
- waist measurement
- shoulder width

using landmark coordinate distances.

Step 4 — Body Position and Angle Detection

The system analyses:

- body posture
- body tilt
- shoulder alignment
- body orientation angle

using AI skeletal tracking.

Step 5 — CSV Dataset Processing

The clothing dataset contains:

- brand names
- garment types
- clothing sizes

- chest measurements
- waist measurements
- fit categories

The dataset is processed using Pandas.

Step 6 — Outfit Recommendation

The AI system compares:

- user body measurements
with:
- clothing dataset measurements

to recommend:

- suitable outfit size
 - clothing category
 - fit type
-

6. Code Implementation:

```
1  import cv2
2  import mediapipe as mp
3  import pandas as pd
4  import math
5
6  height = float(input("Enter Height (cm): "))
7  fit = input("Preferred Fit (Slim/Regular/Loose): ")
8
9  data = pd.read_csv("fashion_brand_measurements_dataset")
10
11 image = cv2.imread("woman.jpg")
12
13 if image is None:
14     print("Image not found")
15     exit()
16
17 image = cv2.resize(image, (720, 720))
18
19 mp_pose = mp.solutions.pose
20 mp_draw = mp.solutions.drawing_utils
21
22 pose = mp_pose.Pose(
23     static_image_mode=True,
24     min_detection_confidence=0.7
25 )
```

```

rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

results = pose.process(rgb)

if results.pose_landmarks:

    lm = results.pose_landmarks.landmark

    h, w, _ = image.shape

    ls = lm[11]
    rs = lm[12]

    x1, y1 = int(ls.x*w), int(ls.y*h)
    x2, y2 = int(rs.x*w), int(rs.y*h)

    shoulder_pixels = math.sqrt(
        (x2-x1)**2 + (y2-y1)**2
    )

    scale = height / h

    shoulder = shoulder_pixels * scale
    chest = shoulder * 1.8
    waist = chest * 0.85

```

```

angle = int(
    math.degrees(
        math.atan2(y2-y1, x2-x1)
    )
)

if abs(y1-y2) < 20:
    position = "Straight"
elif y1 > y2:
    position = "Tilted Right"
else:
    position = "Tilted Left"

match = data[
    (data['chest_cm'] >= chest-5) &
    (data['waist_cm'] >= waist-5) &
    (data['fit_type'].str.lower() == fit.lower())
]

recommendation = "No Match"

if not match.empty:

    best = match.iloc[0]

```

```

best = match.iloc[0]

recommendation = (
    f"{best['brand']} | "
    f"{best['garment_type']} | "
    f"{best['size']}"
)

mp_draw.draw_landmarks(
    image,
    results.pose_landmarks,
    mp_pose.POSE_CONNECTIONS
)

cv2.putText(
    image,
    f"Chest: {chest:.1f} cm",
    (20,40),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    (0,255,255),
    2
)

```

```

cv2.putText(
    image,
    f"Waist: {waist:.1f} cm",
    (20,80),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    (0,255,255),
    2
)

cv2.putText(
    image,
    f"Angle: {angle}",
    (20,120),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    (255,255,0),
    2
)

cv2.putText(
    image,
    f"Position: {position}",
    (20,160),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    (255,255,0),
    2
)

```

```

        0.7,
        (255,255,0),
        2
    )

    cv2.putText(
        image,
        recommendation,
        (20,200),
        cv2.FONT_HERSHEY_SIMPLEX,
        0.6,
        (0,255,0),
        2
    )

    print("Recommended:", recommendation)

else:
    print("No Human Detected")

cv2.imwrite("output.jpg", image)

cv2.imshow("Fashion Recommendation", image)

cv2.waitKey(0)

cv2.destroyAllWindows()

```

Output:



7 Dataset Information

The CSV dataset contains:

- clothing brand names
- garment categories
- chest measurements
- waist measurements
- clothing sizes
- fit types

Example dataset fields:

Field	Description
brand	Clothing brand
garment_type	Type of clothing
size	Clothing size
chest_cm	Chest measurement
waist_cm	Waist measurement
fit_type	Clothing fitting style

8. Applications

The system can be used in:

- virtual fitting rooms
- online shopping systems
- AI fashion assistants
- intelligent clothing recommendation
- smart retail systems
- digital fashion platforms

9. Advantages

The advantages include:

- improved shopping experience
 - intelligent outfit recommendation
 - AI body measurement estimation
 - automatic clothing size prediction
 - reduced clothing return rates
 - better fitting analysis
-

10. Challenges Faced

Challenge	Impact
Poor lighting	Reduces landmark accuracy
Side body pose	Affects measurement estimation
Loose clothing	Reduces body visibility
Complex background	Reduces detection accuracy

11. Expected Output

The AI system generates:

- skeletal tracking
 - body landmark visualization
 - chest and waist estimation
 - body angle detection
 - body position analysis
 - outfit recommendation
 - clothing size prediction
-

12. Learning Outcomes

During this project, the following concepts were understood:

- human pose estimation
- AI landmark detection
- body measurement analysis

- body angle calculation
 - CSV dataset processing
 - AI fashion recommendation
 - computer vision workflows
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13. Conclusion

The AI-Based Fashion Measurement and Outfit Recommendation System successfully demonstrated the integration of:

- Artificial Intelligence
- Computer Vision
- Pose Estimation
- Dataset Processing
- Intelligent Fashion Recommendation

The project improved understanding of:

- AI body tracking
- measurement estimation
- skeletal visualization
- outfit recommendation systems
- virtual fitting technologies

This project demonstrates how AI can improve digital fashion systems and intelligent shopping platforms using body analysis and dataset-driven recommendation logic.